SIMATS ENGINEERING

ASSESSMENT TOOL 2

Case study

COURSE NAME: Computer Networks **COURSE CODE:** CSA0702

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Code of Conduct:

I SHAIK KHAJAH ROSHAN 192511445 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Shaik Khajah

20/20

QUESTIONS:4 **Total Marks:40**

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss, 180 ms jitter,** and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer.**
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms,** significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS, DNS pre-fetching,** and **TTL optimization** to reduce the lookup time below **50 ms.**
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values, Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control, Nagle's algorithm**, and **SYN queue limitations.**
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q1 → 1

The problem is mainly caused by both the UDP transport protocol and the application layer. video conferencing applications use UDP because it provides low latency and does not retransmit lost packets. However, the reported 8% Packet loss. 180ms jitter, and fluctuating bandwidth cause voice distortion, frozen video, and delayed communication, if the application's jitter buffer is not properly configured, these network issues become more noticeable.

2, Real-time multimedia applications require:

- low latency for smooth communication.
- low jitter for continuous audio and video playback.
- Minimal packet loss to maintain quality.
- Fast data delivery without retransmission delays.

UDP is preferred because it sends data quickly without waiting for acknowledgments. making voice and video communication less smooth.

3, Protocol-level improvements:

- Continue using UDP with RTP (Real-time Transport Protocol).
- Enable quality of service (QOS) to prioritize conferencing traffic.
- Use forward error correction (FEC) to reduce the impact of packet loss.

Application-level improvements:

- Increase the adaptive jitter buffer.
- Use adaptive bitrate streaming based on available bandwidth.
- Apply efficient codes such as OPUS (audio) and H.264/H.265 (Video).
- Implement packet-loss concealment techniques.

Question - 2

1. Analyze the reasons behind the high DNS resolution time.

The high DNS resolution time is caused by several factors:

- Centralized DNS servers create a bottleneck when handling many requests.
- Long distance between users and DNS servers increases network latency.
- Lack of DNS caching forces repeated look ups for the same domain names.
- Improper TTL (Time-To-Live) values cause frequent DNS queries.
- Network congestion and overloaded DNS servers delay responses.

2) A suitable DNS architecture should include:

- Anycast DNS: Deploy multiple DNS servers worldwise using the same IP address.
- DNS - Pre-fetching: Browsers resolve frequently used domain names in advance, reducing waiting time.
- TTL optimization: Set moderate TTL values (e.g 300-600 seconds) so clients can timely updates.

• Distributed DNS servers: Place DNS servers in multiple geographic regions for load balance and fault tolerance.

• Local DNS caching: Use ISP and browser caches to avoid repeated DNS queries.

3, Advantages:

• Faster DNS resolution with Anycast routing.

• Reducing latency and quicker webpage loading

• High availability because traffic is redirec-ted if one server fails.

• Better Scalability by distributing traffic across multiple servers.

Limitations:

• Higher deployment and maintenance cost.

• More complex configuration and monitoring.

• DNS cache may temporarily contain outdated records if TTL is too long.

• Anycast routing requires proper network management.

Q. 3.

4. A centralized DNS requests goto a single Server, Creating a bottleneck.

- High traffic increases DNS lookup time, causing slow application loading.
- A failure of the primary DNS Server can make the application unavailable.
- Increased latency affects user experience, especially for global users.
- Limited scalability makes it difficult to handle millions of DNS requests during peak traffic.

5, A distributed DNS servers solution should include:

- Deploy multiple DNS servers in different geographic regions.
- Use Anycast DNS so users automatically connect to the nearest DNS server.
- Enable DNS catching at local DNS resolvers and broarsers.

- Maintain primary and secondray DNS Servers evently.
- Use health monitoring to redirect traffic if a DNS Server fails.

6. TTL (Time-To-Live):

- Controls how long DNS records are stored in cache.
- Proper TTL values reduces repeated DNS queries and improve response time.
- Very long TTL values may delay updates, while very short TTL values increase Server load.

Anycast Routing:

- Directs users to the nearest available DNS server.

DNS caching:

- Stores previously resolved DNS records.
- Reduces DNS lookup time and network traffic.

Q.4.

i, Analyzes.

a) Flow control

- TCP flow control limits the amount of data data sent based on the receivers buffer.

- During heavy traffic, receiver buffers become full, slowing down data transmission and increasing latency.

b, Nagle's Algorithm

- Nagle's algorithm combines small packets before sending them.

- This reduces network overhead but introduces delays for small, time-sensitive trading messages.

c, SYN Queue limitations

- During peak trading hours, many clients try to establish TCP connections simultaneou-sly.

- If the SYN queue becomes full, connection request are delayed or dropped.

2)

- Flow control: Slows data transfer when buffers are full, delaying trade confirmations.
- Nagle's Algorithm: Delays small order packets.
- SYN Queue Limitations: Delays new TCP connections, causing customers to wait longer for order processing

3)

- Disable Nagle's Algorithm using the TCP_NODELAY option to send small packets.
- Increase TCP buffer sizes to handle heavy traffic more efficiently.
- Increase the SYN queue size to support more simultaneous connection requests.
- Enable TCP Fast open (TFO) to reduce connection establishment time.
- Tune TCP retransmission setting to reduce unnecessary retransmissions.

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